Project Title
Patent Prior-Art & Litigation-Analytics Engine

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The Patent Prior-Art & Litigation-Analytics Engine is an AI system that assists patent examiners, researchers, and litigators in identifying relevant prior art and assessing patent validity. It searches large patent corpora using information retrieval and natural language processing to find documents with similar technical concepts and claim elements. The system generates similarity scores for claims, analyzes citation networks to map relationships among prior-art references, and highlights influential patents. It also computes a minimum invalidating set of references, identifying the smallest combination that collectively covers all elements of a target claim. Additionally, it supports examiner assignment by matching applications to examiners based on technical expertise and workload balance. By integrating claim similarity analysis, citation-network analytics, optimization methods, and assignment modeling, the engine reduces manual search effort, improves prior-art discovery accuracy, and provides data-driven support for patent examination and litigation decisions across complex technological domains and large-scale patent datasets worldwide efficiently reliably.

Signature of the Guide